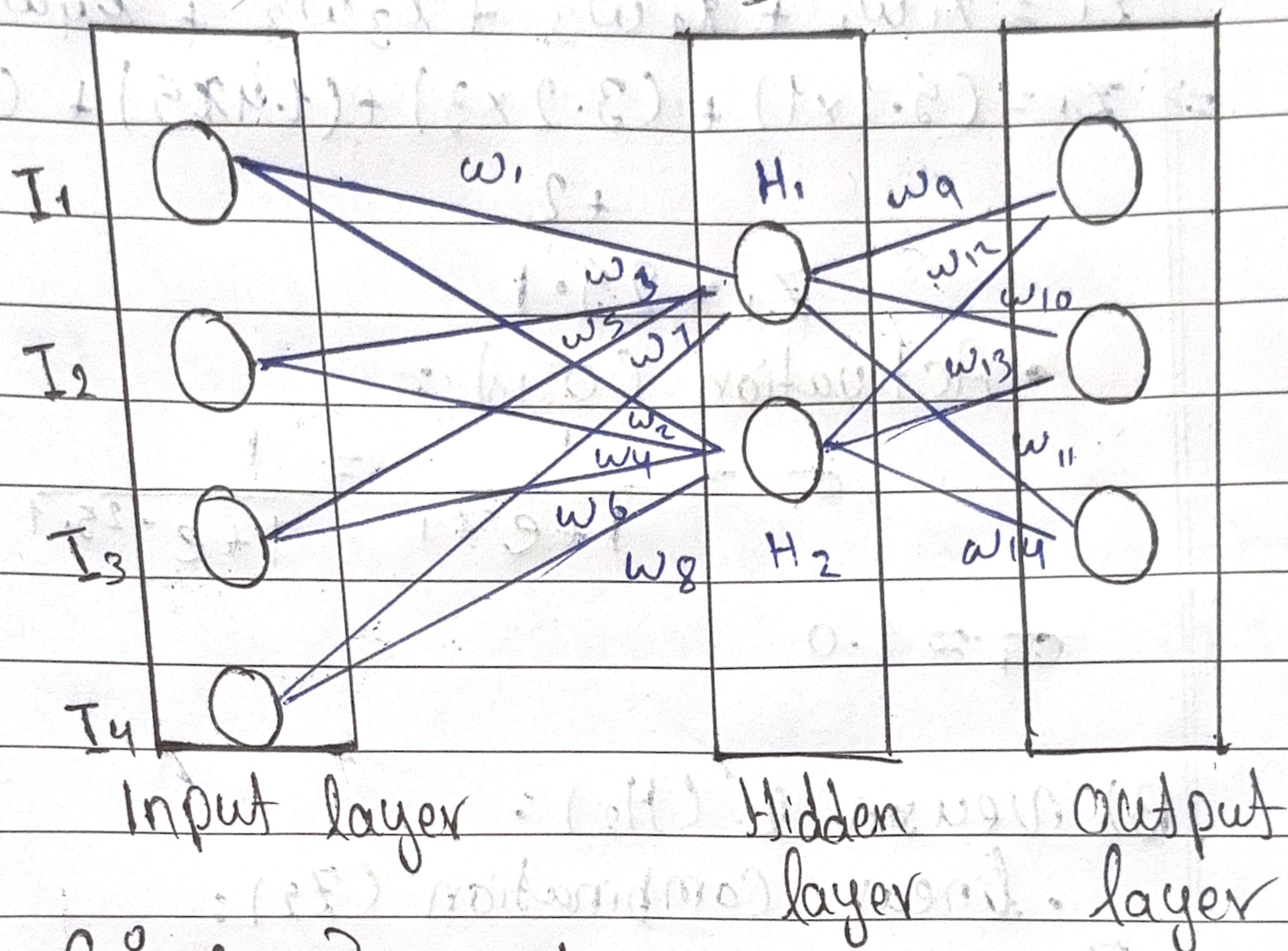


Mathematical Calculation

Single - Hidden - Layer

Neural Network



1. Given Parameters

- Input vector: $X = \{5.1, 3.2, 1.4, 0.2\}$
- Hidden weights (H_1): $w_1 = 1$ $w_3 = 3$
 $w_5 = 5$ $w_7 = 7$
- Hidden weights (H_2): $w_2 = 2$ $w_4 = 4$
 $w_6 = 6$ $w_8 = 8$
- Output weights (from H_1): $w_9 = 1.3$,
 $w_{10} = 0.5$ $w_{11} = 1.2$
- Output weights (from H_2): $w_{12} = 0.5$
 $w_{13} = 0.2$ $w_{14} = 0.4$
- Biases: $b_1 = 2$ $b_2 = 0.1$

2. Hidden layers Calculation

A) Neuron 1 (H_1):

- Linear Combination (Z_1):

$$Z_1 = x_1 w_1 + x_2 w_2 + x_3 w_3 + x_4 w_4 + b_1$$

$$\Rightarrow Z_1 = (5.1 \times 1) + (3.2 \times 3) + (1.4 \times 5) + (0.2 \times 7)$$

$$Z_1 = 25.1$$

- Activation (σ_1):

$$\sigma_1 = \frac{1}{1 + e^{-Z_1}} = \frac{1}{1 + e^{-25.1}}$$

$$\sigma_1 \approx 1.0$$

B) Neuron 2 (H_2):

- Linear Combination (Z_2):

$$Z_2 = x_1 w_5 + x_2 w_6 + x_3 w_7 + x_4 w_8 + b_2$$

$$Z_2 = (5.1 \times 2) + (3.2 \times 4) + (1.4 \times 6) + (0.2 \times 8) + 2.4$$

$$Z_2 = 35.0$$

$$Z_2 = 35.0$$

- Activation (σ_2):

$$\sigma_2 = \frac{1}{1 + e^{-Z_2}} = \frac{1}{1 + e^{-35.0}}$$

$$\sigma_2 \approx 1.0$$

3. output layer calculation

(Combining both neurons)

A) Petosa (Z_{setosa}):

$$Z_{setosa} = (\sigma_1 \times w_9)$$

$$= (1.0 \times 1)$$

$$Z_{setosa} = 1$$

B) Versicolour

$$Z_{versicolour} =$$

$\Rightarrow$

Z_{ver}

C) Virginica

$Z_{virginica}$

Z_{via}

4. final Prediction

- output v

- Prediction

3. output layer calculation [Combining both neurons]

A) Setosa (Z_{setosa}):

$$\begin{aligned} Z_{setosa} &= (\sigma_1 \times w_{10}) + (\sigma_2 \times w_{12}) + b_2 \\ &= (1.0 \times 1.3) + (1.0 \times 0.5) + 0.1 \\ Z_{setosa} &= 1.9 \end{aligned}$$

B) Versicolour ($Z_{versicolor}$)

$$\begin{aligned} Z_{versicolor} &= (\sigma_1 \times w_{10}) + (\sigma_2 \times w_{13}) + b_2 \\ &\Rightarrow (1.0 \times 0.5) + (1.0 \times 0.2) + 0.1 \\ &= 0.5 + 0.2 + 0.1 \\ Z_{versicolor} &= 0.8 \end{aligned}$$

C) Virginica ($Z_{virginica}$)

$$\begin{aligned} Z_{virginica} &= (\sigma_1 \times w_{11}) + (\sigma_2 \times w_{14}) + b_2 \\ &\Rightarrow (1.0 \times 1.2) + (1.0 \times 0.4) + 0.1 \\ Z_{virginica} &= 1.7 \end{aligned}$$

4. final Prediction

- output vector: $[1.9, 0.8, 1.7]$
- Prediction: Setosa [Highest, 1.9]